The Krenn–Gu conjecture holds for all simple graphs on six vertices

Junhao Liang
Zhejiang Normal University
mailto:JunhaoLiang2005@gmail.com

Abstract

We prove that for *every* number of colors $D \geq 3$ there is no bi-colored weighted *simple* graph on 6 vertices — bichromatic edges and complex weights allowed — in which all D monochromatic inherited vertex colorings have unit weight while every other vertex coloring has weight zero. Equivalently, the matching index of every simple graph on 6 vertices is at most 2; K_4 (with $\mu = 3$) remains the only known exception to $\mu(G) \leq 2$ at this order. For $D = 3, 4, 5$ the proof reduces the statement to a purely combinatorial finite search: a complete enumeration of the 5,610, 2,160 and 720 possible monochromatic-class configurations respectively, collapsed to 24, 5 and 1 orbit representatives under the symmetric group S_6, followed by an exhaustive check of all 2,033,610, 4,917 and 1 edge-color assignments — every one of which is eliminated by one of two elementary necessary conditions; for $D \geq 6$ the statement follows from a one-line edge-counting argument ($3D \leq 15$). Because the elimination is weight-free, the result holds over any integral domain, in particular over $\mathbb{C}$, $\mathbb{R}$, $\mathbb{Z}$, and for weights in $\{-1, 0, 1\}$. The certificate is machine-reproducible (seconds of computation), has been independently re-verified by fresh implementations and by an independent SAT encoding, and settles the open statements `eqSystem6_no_solution_d3`, `_d4`, `_d5` and `_ge3` (with their `_real`, `_int`, `_trinary_int` variants where present) in the DeepMind *formal-conjectures* benchmark. The full conjecture (arbitrary $n \geq 8$, multigraphs) remains open.

1 Introduction

The following conjecture emerged from quantum optics in 2017 [1] and was stated in graph-theoretic form by Krenn, Gu and Soltész [2]:

> *If G is a graph with more than 4 vertices, then the matching index satisfies $\mu(G) \leq 2$.*

Here $\mu(G)$ is the dimension of the subspace spanned by "monochromatic" perfect-matchings superpositions; the complete graph K_4 attains $\mu(K_4) = 3$ and is conjecturally the only graph of matching index at least 3 (see [2, 4] for the exact definitions and the prize page [3] for the current status and the prize associated with a full resolution). The conjecture has survived substantial computational effort: SAT-based searches ruled out mono-edge constructions for small orders [7], and the best partial results show $\mu(G) \leq 2$ for graphs of vertex connectivity at most 2 and for cubic graphs, with any minimal counterexample forced to be 4-connected [6]. For *simple* graphs, the strongest general bound is $\mu(G) \leq n/\sqrt{2}$ (Theorem 22 of [5]); at $n = 6$ this yields only $\mu(G) \leq 4$, since μ is integral, whereas the present paper proves $\mu(G) \leq 2$. Over *nonnegative* real weights the conjecture is fully settled (Bogdanov, see [8]), and the DeepMind formal-conjectures benchmark records the general case as open, with only the $D = N$ cases resolved formally [9, 10].

In this note we resolve the case $n = 6$ *completely for simple graphs*: the conjecture holds for every number of colors $d \geq 3$. Our main result is:

Theorem 1. *For every $D \geq 3$ there is no bi-colored weighted simple graph on 6 vertices — with each edge carrying an ordered color pair in $\{0, \ldots, D-1\}^2$ and a nonzero weight in an integral domain R — such that $w(\sigma) = 1$ for every monochromatic coloring $\sigma = i^6$, $i \in \{0, \ldots, D-1\}$, and $w(\sigma) = 0$ for every other vertex coloring.*

Consequently $\mu(G) \leq 2$ for every simple graph G on 6 vertices (over any integral domain). Two features deserve emphasis:

(a) The obstruction is *purely combinatorial*: the necessary conditions (b) and (c) of Proposition 7 below never mention weights. Hence the result is independent of the coefficient domain.

(b) The certificate is *exhaustive and machine-checkable*: the entire search space (2,033,610 color assignments) is enumerated and eliminated in seconds, with no reliance on symbolic or numeric solvers.

The result settles the DeepMind formal-conjectures open statements `eqSystem6_no_solution_d3`, `_d4`, `_d5` and the universal `_ge3` family, with their `_real`, `_int` and `_trinary_int` variants where present (Section 6).

2 Preliminaries

We follow the terminology of [2, 6]. Fix the vertex set $V = \{0, 1, 2, 3, 4, 5\}$. A *bi-colored weighted graph* is a triple (V, E, c, w) where $E \subseteq \binom{V}{2}$ is a simple graph, each edge $e = \{u, v\} \in E$ $(u < v)$ carries an ordered color pair $c(e) = (c_1, c_2) \in \{0, 1, 2\}^2$ (c_1 at u, c_2 at v), and $w : E \to R$ assigns to each edge a nonzero weight in a commutative integral domain R. A *perfect matching* (PM) of (V, E) is a set of three pairwise disjoint edges covering all six vertices. For a PM M, the *inherited vertex coloring* $\mathrm{IVC}(M) \in \{0, 1, 2\}^6$ is defined by giving each vertex the color of its incident matching edge at that vertex; for $\sigma \in \{0, 1, 2\}^6$ put

$$w(\sigma) \;=\; \sum_{M:\ \mathrm{IVC}(M)=\sigma} \; \prod_{e \in M} w(e).$$

A $(6, 3)$-*solution* is a bi-colored weighted graph with $w(000000) = w(111111) = w(222222) = 1$ and $w(\sigma) = 0$ for all other σ.

The fifteen PMs of K_6 are the candidates; a PM M of K_6 *survives* in (V, E, c, w) if $M \subseteq E$.

3 Reduction to a finite combinatorial search

The following lemmas are elementary; they distill the argument of [2] and the project notes [11]. All of them hold over any integral domain.

Lemma 2 (All-nonzero WLOG)**.** *If a $(6, 3)$-solution exists, one exists with all weights nonzero.*

Proof. Delete every edge of weight 0: such edges contribute 0 to every surviving PM product, so every sum $w(\sigma)$ is unchanged, and the monochromatic sums remain 1, so each color still has a surviving PM. ☐

Lemma 3 (K_6 suffices)**.** *If some simple graph on* 6 *vertices has a* $(6,3)$*-solution, then so does* K_6 *(with missing edges colored arbitrarily and given weight* 0*).*

Proof. Extend the coloring of E to all fifteen pairs arbitrarily and set the weight of every new edge to 0. Every new PM uses a zero-weight edge and contributes nothing. ☐

By Lemmas 2 and 3, the search space is: color each of the fifteen pairs of V by an ordered pair in $\{0,1,2\}^2$ or mark it *absent* (weight 0), and keep only colorings in which every surviving PM has all nonzero weights — which, by Lemma 2, every surviving solution may be assumed to satisfy.

Lemma 4 (Monochromatic PMs)**.** *For a PM* M *with all edges present,* $\mathrm{IVC}(M) = iiiiii$ *(write* i^6*) iff every edge of* M *has color pair* (i,i)*.*

Proof. Vertex u receives color i iff its matching edge has color i at u. If all six vertices receive color i, both endpoint colors of every matching edge equal i. ☐

Write P_i for the set of surviving PMs with IVC i^6; by Lemma 4, $P_i = \mathrm{PM}(H_i)$ where H_i is the subgraph of color-(i,i) edges. An edge has a single color pair, so H_0, H_1, H_2 are pairwise edge-disjoint.

Lemma 5 (Per-IVC decomposition)**.** $w(\sigma)$ *depends only on the sets* P_i *and the non-monochromatic IVC groups:* $w(i^6) = \sum_{M\in P_i}\prod_{e\in M} w(e)$*, and* $w(\sigma) = \sum_{M:\,\mathrm{IVC}(M)=\sigma}\prod_{e\in M} w(e)$ *for non-monochromatic* σ*.*

Lemma 6 (The L3 condition)**.** *In a solution with all weights nonzero, every* realized *non-monochromatic IVC is realized by at least two surviving PMs.*

Proof. If a unique surviving PM M realizes σ, then $w(\sigma) = \prod_{e\in M} w(e) \neq 0$ (nonzero product in an integral domain), contradicting $w(\sigma) = 0$. ☐

Proposition 7 (Necessary conditions)**.** *Every* $(6,3)$*-solution induces a coloring of the fifteen pairs of* V *— each pair colored by a pair in* $\{0,1,2\}^2$ *or absent — satisfying*

(a) P_0, P_1, P_2 *are all nonempty (Lemma 5 with* $w(i^6) = 1$*),*

(b) no surviving PM outside $P_0 \cup P_1 \cup P_2$ *has a monochromatic IVC (Lemma 4),*

(c) every realized non-monochromatic IVC is realized by at least two surviving PMs (Lemma 6).

It is (a)–(c) — and nothing else — that the search below eliminates. Note that no condition on the coefficient domain R was used anywhere; the whole argument is weight-free.

4 The exhaustive certificate

4.1 Class enumeration and orbit reduction

By (a) and Lemma 4, a surviving solution determines an ordered triple (P_0, P_1, P_2) of *nonempty, pairwise edge-disjoint* subsets of the set of the fifteen PMs of K_6. A direct construction enumerates all such triples:

Proposition 8 (Class count, computational)**.** *There are exactly* 5,610 *ordered triples* (P_0, P_1, P_2) *of nonempty, pairwise edge-disjoint PM subsets of* K_6*.*

The symmetric group S_6 acts on vertex labels; we fix the three color labels throughout (no color renaming is used). Canonicalizing each triple under this action collapses the 5,610 classes to 24 orbit representatives, listed in Appendix A. The completeness statement is:

Proposition 9 (Orbit completeness, computational)**.** *For every one of the* 5,610 *triples of Proposition 8, some permutation in S_6 sends it to one of the* 24 *representatives in Appendix A; the* 24 *representatives are pairwise inequivalent.*

4.2 The batch

Fix an orbit representative (P_0, P_1, P_2). The edges appearing in the PMs of $P_0 \cup P_1 \cup P_2$ are *forced*: an edge of a PM of P_i must carry the color pair (i, i) (Lemma 4). Every remaining (*free*) edge may be colored by any of the 9 ordered pairs in $\{0, 1, 2\}^2$ or be absent. Writing F for the number of free edges of the representative, the batch enumerates all 10^F assignments. For every assignment we compute the IVC code of each of the fifteen PMs of K_6 (a PM using an absent edge is *dead* and contributes nothing) and test the two conditions:

no-new-mono (b): no PM outside its class (a dead PM, or a PM of P_j for $j \neq i$, or an unclassified PM) has IVC i^6 for any $i \in \{0, 1, 2\}$;

L3 (c): among the *realized* non-monochromatic IVC codes, no code occurs exactly once.

A coloring that passes both tests would proceed to the exact decision procedure (linear system over $\mathbb{Q}$ together with the liftability relations $\ker U^{\mathsf{T}}$); see [11]. This stage is never reached:

Theorem 10 (Exhaustive elimination, computational)**.** *For each of the* 24 *representatives of Appendix A, every one of the* 10^F *assignments fails (b) or (c). In total,* 2,033,610 *assignments are eliminated; the number of survivors is* 0*.*

Table 1 records the distribution of the representatives by class sizes and free-edge count; the total is $\sum 10^F = 2{,}033{,}610$.

Proof of Theorem 1. Suppose a $(6, 3)$-solution exists over an integral domain R. By Lemmas 2–3 it induces a coloring of the fifteen pairs (each pair colored or absent) with all necessary weights nonzero, whose triple (P_0, P_1, P_2) is one of the 5,610 classes. By Proposition 9 some S_6-image of this coloring has the same triple as one of the 24 representatives, and the induced free- edge assignment is one of the 10^F enumerated. By Proposition 7 the coloring satisfies (a)–(c); but Theorem 10 says no enumerated assignment satisfies (b) and (c) — contradiction. □

4.3 Four and five colors, and the case $D \geq 6$

The method extends verbatim to $D = 4, 5$ colors. A solution with D colors determines D pairwise edge-disjoint nonempty monochromatic PM classes, and the edge-counting argument below already bounds the possibilities:

Lemma 11 (Only five colors can occur)**.** *In a $(6, D)$-solution over any integral domain, $D \leq 5$.*

Proof. For each color i the monochromatic sum $w(i^6) = 1$ forces the class P_i to be nonempty (an empty sum is $0 \neq 1$). The classes are pairwise edge-disjoint (an edge carries one color pair), and each nonempty class contains a PM, hence at least 3 edges. Therefore $3D \leq 15$. □

class size $(\lvert P_0\rvert, \lvert P_1\rvert, \lvert P_2\rvert)$	free edges F	multiplicity
$(1, 1, 1)$	6	2
$(1, 1, 2)$	$3, 4$	$1, 1$
$(1, 1, 3)$	$0, 2$	$1, 1$
$(1, 1, 4)$	0	1
$(1, 2, 1)$	$4, 3$	$1, 1$
$(1, 2, 2)$	$2, 0$	$1, 1$
$(1, 3, 1)$	$2, 0$	$1, 1$
$(1, 4, 1)$	0	1
$(2, 1, 1)$	$4, 3$	$1, 1$
$(2, 1, 2)$	$2, 0$	$1, 1$
$(2, 2, 1)$	$2, 0$	$1, 1$
$(2, 2, 2)$	0	1
$(3, 1, 1)$	$2, 0$	$1, 1$
$(4, 1, 1)$	0	1

Table 1: The 24 orbit representatives: class sizes, free-edge counts, and multiplicities. Total rows: $2 \cdot 10^6 + 10^4 \cdot 2 + 10^3 \cdot 2 + 10^2 \cdot 4 + 1 \cdot 8 = 2{,}033{,}610$.

For $D = 4$: enumerating all ordered quadruples of nonempty pairwise edge-disjoint PM subsets yields 2,160 classes, which collapse to 5 orbit representatives under S_6 (Appendix A); every free edge has $16 + 1$ options, the free-edge count satisfies $F \leq 15 - 12 = 3$, and the complete batch enumerates 4,917 assignments, all eliminated by (b) or (c).

For $D = 5$: an ordered quintuple forces all five classes to be single PMs (two PMs in one class would force the union to contain $5 + 12 = 17 > 15$ edges), i.e. the classes form an ordered 1-factorization of K_6; there are exactly $6 \cdot 5! = 720$ such quintuples, all equivalent under S_6 — one orbit, one forced coloring ($F = 0$), eliminated by (b) or (c).

Combined with Lemma 11, the three exhaustive searches settle Theorem 1 in full.

5 Independent verification

Because the theorem rests on a computation, the computation was re-implemented from scratch and cross-validated by independent means. All artifacts (scripts, logs, this certificate) accompany the paper [12].

(V1) *Fresh implementation.* The PM enumeration, the triple enumeration, the canonicalization, the IVC computation, and the filter were rewritten independently (no shared decision logic), reproducing 5,610 and the 24-orbit completeness claim, and re-running the full batch: 2,033,610 rows, 0 survivors.

(V2) *Filter validation.* The vectorized filter was cross-checked against a naive per-row reference implementation on 20,000 random rows, and the original implementation's filter was cross-checked against the same reference on 60,000 random rows; full agreement.

(V3) *SAT confirmation.* An independent encoding of the constraints (b) and (c) for each of the 24 classes was solved with the SMT solver z3: `UNSAT` in all 24 cases.

(V4) *Decision-procedure controls.* A second, independently written exact decision engine agrees with the original engine on 300 random colorings and 200 structured instances (100% agreement), and passes positive controls: the known 3-dimensional example on 4 vertices (the K_4 witness) is correctly certified feasible with an exact weight construction, and relaxed instances exercise the lift-relation machinery.

(V5) *Mono-edge subcase.* Of the $3^{15} = 14{,}348{,}907$ mono-edge colorings, exactly 248,310 realize all three monochromatic IVCs, and 0 of them satisfy (c) — recovering and strengthening the SAT result of [7] for $n = 6$.

(V6) *Four and five colors.* The $D = 4$ and $D = 5$ enumerations (2,160 and 720 classes; 5 and 1 orbits; 4,917 and 1 rows; 0 survivors) were re-derived from scratch: the raw counts by an independent formula ($\sum_T (2^{c(T)} - 1)$ over the audited triple list, resp. $6 \cdot 5!$), the orbit cover by re-canonicalizing every raw class, the batches replayed with an independent naive filter implementation, and the constraints confirmed by z3 (`UNSAT` 5/5 and 1/1).

No randomness is involved in any infeasibility verdict: every elimination in Theorem 10 is a structural check of condition (b) or (c).

6 Consequences for the formal-conjectures benchmark

The DeepMind *formal-conjectures* benchmark [10, 9] contains the open Lean statements $\neg\exists W :$ WeightsN 6 D R, EqSystemN 6 D W for various color counts D and coefficient domains R. Since Theorem 1 holds over any integral domain, it settles the following 13 statements currently recorded as open in the benchmark:

`eqSystem6_no_solution_d3`, `_d3_real`, `_d3_int`, `_d3_trinary_int`,
`eqSystem6_no_solution_d4`, `eqSystem6_no_solution_d5`, `_d5_real`, `_d5_int`, `_d5_trinary_int`,
`eqSystem6_no_solution_ge3`, `_ge3_real`, `_ge3_int`, `_ge3_trinary_int`.

(The benchmark also records `eqSystem6_no_solution_d6` as solved; it follows elementarily from Lemma 11 as well.)

7 Conclusion and open problems

We have settled the Krenn–Gu conjecture for all simple graphs on 6 vertices and all color counts $D \geq 3$. The following directions remain:

(i) *Multigraphs.* The conjecture allows parallel edges. The known reductions merge same-pair parallels and delete self-cancelling families; whether *distinct-pair* parallel edges can be eliminated (thereby extending Theorem 1 to multigraphs on 6 vertices) is open. Computational evidence (restricted layers, cascade analysis) is recorded in the accompanying repository.

(ii) $n \geq 8$. The combinatorial bottleneck (condition (c)) is exactly what fails to generalize naively; new ideas are needed.

(iii) *Formalization.* A Lean formalization of the three enumerations (the 24-, 5- and 1-orbit searches and their 2,033,610-, 4,917- and 1-row eliminations, executable under the Lean kernel) would turn the 13 benchmark statements of Section 6 into `answer(True)` entries.

Remark 12 (Relation to the prize). *The prize of €3,000 [3] is attached to a full resolution of the conjecture (all even $n \geq 6$, multigraphs included). This note does not claim it.*

Acknowledgments

This research was conducted with substantial assistance from AI systems (Anthropic's Claude), which contributed to the design of the exhaustive search, the drafting of the manuscript, and the independent verification described in Section 5. All mathematical claims were reviewed by the author, and every computational claim was re-derived by at least one independent implementation or independent solver. No AI system is a co-author of this paper.

A The 24 orbit representatives

Vertices are $0, \dots, 5$; each PM is written as three edges, e.g. 01-23-45 denotes $\{\{0,1\},\{2,3\},\{4,5\}\}$. Representative number k has classes (P_0^k, P_1^k, P_2^k).

k	P_0^k	P_1^k	P_2^k
0	01-23-45	02-14-35	03-15-24
1	01-23-45	02-14-35	03-15-24 / 04-13-25
2	01-23-45	02-14-35	03-15-24 / 04-13-25 / 05-12-34
3	01-23-45	02-14-35	05-13-24
4	01-23-45	02-14-35	03-15-24 / 05-13-24
5	01-23-45	02-14-35	03-15-24 / 04-13-25 / 05-13-24
6	01-23-45	02-14-35	03-15-24 / 04-13-25 / 05-12-34 / 05-13-24
7	01-23-45	02-14-35 / 02-15-34	04-13-25
8	01-23-45	02-14-35 / 02-15-34	04-13-25 / 05-13-24
9	01-23-45	02-15-34 / 03-14-25	04-12-35
10	01-23-45	02-15-34 / 03-14-25	04-12-35 / 05-13-24
11	01-23-45	02-14-35 / 02-15-34 / 03-14-25	05-13-24
12	01-23-45	02-15-34 / 03-14-25 / 04-12-35	05-13-24
13	01-23-45	02-14-35 / 02-15-34 / 03-14-25 / 04-12-35	05-13-24
14	01-23-45 / 01-24-35	02-15-34	03-14-25
15	01-23-45 / 01-24-35	02-15-34	03-14-25 / 04-13-25
16	01-23-45 / 01-24-35	03-14-25 / 04-13-25	02-15-34
17	01-23-45 / 01-24-35	03-14-25 / 04-13-25	02-15-34 / 05-12-34
18	01-24-35 / 02-13-45	03-14-25	04-15-23
19	01-24-35 / 02-13-45	03-14-25	04-15-23 / 05-12-34
20	01-24-35 / 02-13-45	03-14-25 / 04-15-23	05-12-34
21	01-23-45 / 01-24-35 / 02-13-45	03-14-25	05-12-34
22	01-25-34 / 02-14-35 / 03-12-45	04-15-23	05-13-24
23	01-24-35 / 02-13-45 / 02-14-35 / 03-14-25	04-15-23	05-12-34

For $D = 4$ colors, the 5 orbit representatives (each class P_i^k nonempty, classes pairwise edge-disjoint) are:

k	P_0^k	P_1^k	P_2^k	P_3^k
0	01-23-45	02-14-35	03-15-24	04-13-25
1	01-23-45	02-14-35	03-15-24	04-13-25 / 05-12-34
2	01-23-45	02-14-35	03-15-24 / 04-13-25	05-12-34
3	01-23-45	02-15-34 / 03-14-25	04-12-35	05-13-24
4	01-24-35 / 02-13-45	03-14-25	04-15-23	05-12-34

Representative 0 has $F = 3$ free edges ($4{,}913 = 17^3$ assignments); representatives 1–4 have $F = 0$ (their classes partition all 15 edges); total 4,917 rows.

For $D = 5$ colors the unique orbit representative is the ordered 1-factorization

$$P_0 = \{\text{01-23-45}\}, \quad P_1 = \{\text{02-14-35}\}, \quad P_2 = \{\text{03-15-24}\}, \quad P_3 = \{\text{04-13-25}\}, \quad P_4 = \{\text{05-12-34}\},$$

with $F = 0$: one assignment, eliminated by the conditions.

B Reproducibility

All computations run on a standard Python 3.9 + numpy 2.0 + sympy 1.14 stack (z3 for the SAT cross-check), take a few seconds for the batch, and contain no randomized components on the certification path. Scripts: `computation/gen_search.py` (class enumeration and batch), `computation/search_core.py` (exact decision procedure), and the independent verification suite `audit/verification_2026-09-03/` described in Section 5. The complete audit report is `audit/AUDIT_REPORT_2026-09-03.md`.

References

[1] M. Krenn, X. Gu, A. Zeilinger, *Quantum experiments and graphs: Multiparty states as coherent superpositions of perfect matchings*, Phys. Rev. Lett. 119, 240403 (2017).

[2] M. Krenn, X. Gu, D. Soltész, *Questions on the structure of perfect matchings inspired by quantum physics*, Proc. 2nd Croatian Combinatorial Days (2019), 57–70; arXiv:1902.06023.

[3] M. Krenn, *Inherited vertex coloring of graphs*, bounty page, `https://mariokrenn.wordpress.com/graph-theory-question/` (mirrored verbatim at `https://openproblemgarden.org/op/monochromatic_vertex_colorings_inherited_from_perfect_matchings`)

[4] L. S. Chandran, R. Gajjala, C. Illickan, *Krenn–Gu conjecture for sparse graphs*, MFCS 2024, LIPIcs 306, art. 41; arXiv:2407.00303.

[5] L. S. Chandran, R. Gajjala, *Graph-theoretic insights on the constructability of complex entangled states*, Quantum 8, 1396 (2024), DOI 10.22331/q-2024-07-03-1396; arXiv:2304.06407.

[6] L. S. Chandran, R. Gajjala, *Edge-coloured graphs with only monochromatic perfect matchings and their connection to quantum physics*, Electron. J. Combin. (2026); arXiv:2202.05562.

[7] A. Cervera-Lierta, M. Krenn, A. Aspuru-Guzik, *Design of quantum optical experiments with logic artificial intelligence*, Quantum 6, 836 (2022); arXiv:2109.13273.

[8] MathOverflow 311325, *A question on inherited vertex colorings of perfect matchings* (M. Krenn, answers by I. Bogdanov et al.).

[9] DeepMind, *Advancing mathematics research with AI-driven formal proof search*, arXiv:2605.22763 (2026).

[10] google-deepmind/formal-conjectures, *MonochromaticQuantumGraph* Lean formalizations and open statements, `https://github.com/google-deepmind/formal-conjectures`

[11] *Krenn–Gu research workspace: proof notes and certificate*, accompanying repository, 2026.

[12] *Accompanying repository: scripts, logs, audit report and certificate*, `https://github.com/cho-leung/krenn-gu-n6`.